MATH 4610: Fundamentals of Numerical Analysis Fall 2026

Final Project A

Due: Friday, December 18, 2026, 11:59 pm, on Canvas

General Instructions

This is one of three final projects. Choose one of them. Submit on Canvas (i) a report in PDF, typeset in LaTeX or Word, and (ii) a zip file with your code. All code must be written in MATLAB. Include one script for each problem (`Q1.m`, `Q2.m`, `Q3.m`) that produces the figures and tables of that problem, and a short `README` file. For each problem, the report should contain the derivations, the tables and figures, and a brief discussion of the results.

You may discuss the project with others, but your code and your report must be your own. Cite any sources you use.

Final presentation. Each student will present the project in class during the week of December 7–11, on Monday (Dec. 7), Wednesday (Dec. 9), or Friday (Dec. 11). The schedule will be posted on Canvas. Present the problem, your main results, and your conclusions; the work need not be finished. The written report is due on December 18 and may take into account comments made at the presentation.

Grading

Problem 1: discretization, accuracy, and conditioning in 1D	20
Problem 2: Jacobi, Gauss–Seidel, and conjugate gradient methods in 1D	25
Problem 3: the Poisson equation in 2D	25
Code (correct, readable, reproducible)	10
Written report	10
Final presentation	10
Total	100
Bonus: the SOR method	10

Problem 1: The Poisson Equation in 1D (20 points)

Consider

$$-u''(x) = f(x), \quad 0 < x < 1, \qquad u(0) = u(1) = 0,$$

with $f(x) = \pi^2 \sin(\pi x)$, so that the exact solution is $u(x) = \sin(\pi x)$. Let $h = 1/(N+1)$ and $x_i = ih$, $i = 0, \dots, N+1$. The finite difference scheme is

$$-\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} = f(x_i), \quad i = 1, \dots, N, \qquad u_0 = u_{N+1} = 0.$$

(a) Show that the scheme can be written as $A\mathbf{u} = \mathbf{b}$ with $b_i = f(x_i)$ and

$$A = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix} \in \mathbb{R}^{N\times N}.$$

Use Taylor's theorem to show that the local truncation error is $O(h^2)$.

(b) Prove that A is symmetric positive definite. (Hint: for $\mathbf{v} \neq 0$ and $v_0 = v_{N+1} = 0$, show that $\mathbf{v}^T A\mathbf{v} = h^{-2}\sum_{i=0}^{N}(v_{i+1} - v_i)^2 > 0$.)

(c) Solve $A\mathbf{u} = \mathbf{b}$ with MATLAB's backslash, storing A as a sparse matrix. Let $e_i = u(x_i) - u_i$, with $e_0 = e_{N+1} = 0$, and compute

$$\|e\|_{L^2} \approx \Big(h\sum_{i=1}^{N} e_i^2\Big)^{1/2}, \qquad |e|_{H^1} \approx \Big(h\sum_{i=0}^{N}\Big(\frac{e_{i+1}-e_i}{h}\Big)^2\Big)^{1/2}.$$

Do this for $N+1 = 16, 32, 64, 128, 256, 512$. Make a table of both errors and of the observed order $\log_2\big(E(h)/E(h/2)\big)$, and plot the errors against h on a log-log scale. Compare the observed orders with the truncation error found in (a).

(d) The eigenvalues of A are $\lambda_k = \frac{4}{h^2}\sin^2\big(\frac{k\pi h}{2}\big)$, $k = 1, \ldots, N$. Find $\kappa_2(A)$ and its behavior as $h \to 0$. Check your formula with `cond(full(A))` for $N+1 \leq 256$, and plot $\kappa_2(A)$ against N on a log-log scale.

Problem 2: Iterative Methods in 1D (25 points)

Use the system $A\mathbf{u} = \mathbf{b}$ of Problem 1. In every method, start from $\mathbf{u}^{(0)} = 0$ and stop when

$$\frac{\|\mathbf{b} - A\mathbf{u}^{(k)}\|_2}{\|\mathbf{b}\|_2} < 10^{-8},$$

or when $k = 10^6$; report it if this limit is reached.

(a) Write your own MATLAB functions for the Jacobi method, the Gauss–Seidel method, and the conjugate gradient (CG) method. Do not use MATLAB's built-in iterative solvers. Check each function against the solution of Problem 1 for a small N.

(b) For $N+1 = 16, 32, 64, 128, 256$, make a table of the number of iterations and the computing time of each method. For $N+1 = 64$, plot the relative residual against the iteration number on a semilog scale.

(c) For this matrix, the iteration matrix of the Jacobi method has spectral radius $\rho_J = \cos(\pi h)$, and that of the Gauss–Seidel method has spectral radius $\rho_{GS} = \rho_J^2$. Use these values to predict the number of iterations needed to reduce the error by a factor 10^{-8}, and compare with your table. For CG, compare with the bound

$$\|e^{(k)}\|_A \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \|e^{(0)}\|_A, \qquad \kappa = \kappa_2(A).$$

How does the number of iterations of each method grow with N?

(d) For $N+1=64$, solve the system by CG with tolerances $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$ and compute $\|e\|_{L^2}$ in each case. Below which tolerance does the error stop decreasing? Explain.

Problem 3: The Poisson Equation in 2D (25 points)

Consider

$$-\Delta u = f \quad \text{in } \Omega = (0,1)\times(0,1), \qquad u = 0 \quad \text{on } \partial\Omega,$$

with $f(x,y) = 2\pi^2 \sin(\pi x)\sin(\pi y)$, so that $u(x,y) = \sin(\pi x)\sin(\pi y)$. Let m be the number of interior grid points in each direction, $h = 1/(m+1)$, and use the five-point scheme

$$-\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} - \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{h^2} = f(x_i, y_j), \qquad 1 \le i, j \le m.$$

(a) Number the unknowns row by row. Show that the matrix of the system is

$$A = \frac{1}{h^2}\,(I \otimes T + T \otimes I) \in \mathbb{R}^{m^2 \times m^2}, \qquad T = \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{m\times m}.$$

Build A as a sparse matrix with `kron`. Using the eigenvalues of T from Problem 1(d), show that A is symmetric positive definite. How many nonzero entries does a row of A have at most?

(b) Solve the system with backslash for $m+1 = 16, 32, 64, 128$. With $e_{i,j} = u(x_i, y_j) - u_{i,j}$ and $e = 0$ on the boundary, compute

$$\|e\|_{L^2} \approx \Big(h^2 \sum_{i,j} e_{i,j}^2\Big)^{1/2}, \qquad |e|_{H^1} \approx \Big(h^2 \sum_{i,j} \Big[\Big(\frac{e_{i+1,j} - e_{i,j}}{h}\Big)^2 + \Big(\frac{e_{i,j+1} - e_{i,j}}{h}\Big)^2\Big]\Big)^{1/2}.$$

Make a table of the errors and observed orders as in Problem 1, and plot the computed solution and the error for $m+1 = 32$.

(c) With the stopping rule of Problem 2, solve the system for $m+1 = 16, 32, 64, 128, 256$ by

- your CG function from Problem 2,
- preconditioned CG with incomplete Cholesky (`ichol`) or SSOR as the preconditioner,
- backslash.

Make a table of the numbers of iterations and the computing times. For preconditioned CG, give the time to build the preconditioner separately.

(d) How does the number of CG iterations, with and without the preconditioner, grow with m? Relate your answer to $\kappa_2(A)$, which equals $\kappa_2(T)$. Explain why the diagonal (Jacobi) preconditioner does not reduce the number of iterations for this matrix. Which method would you use for $m+1 = 256$, and why?

Bonus: The SOR Method (10 points)

The SOR method for the system of Problem 2 is

$$u_i^{(k+1)} = (1-\omega)\,u_i^{(k)} + \omega\,\tilde{u}_i^{(k+1)},$$

where $\tilde{u}_i^{(k+1)}$ is the Gauss–Seidel value. Implement it. For $N+1 = 32$ and $N+1 = 64$, plot the number of iterations against ω for $\omega = 1, 1.02, 1.04, \ldots, 1.98$. Compare the best ω you find with the theoretical value $\omega_{\text{opt}} = 2/(1+\sin(\pi h))$, and compare the number of SOR iterations at ω_{opt} with those of Gauss–Seidel and CG.